

DBMS Unit 3 Question Bank — Normalization & Schema Refinement

Database Management Systems (DBMS) · Unit 3 · Normalization & Schema Refinement · 2025

Important and frequently asked questions, grouped by marks.

Part A — Short answer (2 marks)

1. Define Functional Dependencies. Derive or justify the key properties involved.
2. Illustrate Closure of Attributes with a worked example.
3. What is meant by First, Second and Third Normal Form? Discuss its advantages and limitations.
4. Explain Boyce-Codd Normal Form with a neat diagram and a suitable example.
5. Define Lossless Join Decomposition. Derive or justify the key properties involved.
6. Explain Multivalued Dependencies and 4NF with a neat diagram and a suitable example.

Part B — Long answer (10 marks)

1. Illustrate Functional Dependencies with a worked example.
2. Explain Closure of Attributes with a neat diagram and a suitable example.
3. Compare First, Second and Third Normal Form with Functional Dependencies. When would you prefer one over the other?
4. What is meant by Boyce-Codd Normal Form? Discuss its advantages and limitations.
5. Illustrate Lossless Join Decomposition with a worked example.
6. Define Multivalued Dependencies and 4NF. Derive or justify the key properties involved.
7. Discuss the role of Functional Dependencies in Normalization & Schema Refinement.
8. Explain Closure of Attributes with a neat diagram and a suitable example.
9. Define First, Second and Third Normal Form. Derive or justify the key properties involved.
10. What is meant by Boyce-Codd Normal Form? Discuss its advantages and limitations.
11. Write short notes on Lossless Join Decomposition.
12. Discuss the role of Multivalued Dependencies and 4NF in Normalization & Schema Refinement.